

Ex No: 1 Butterworth IIR Filters

1.1 Design and Analysis of a Butterworth IIR Low-Pass Filter Using MATLAB

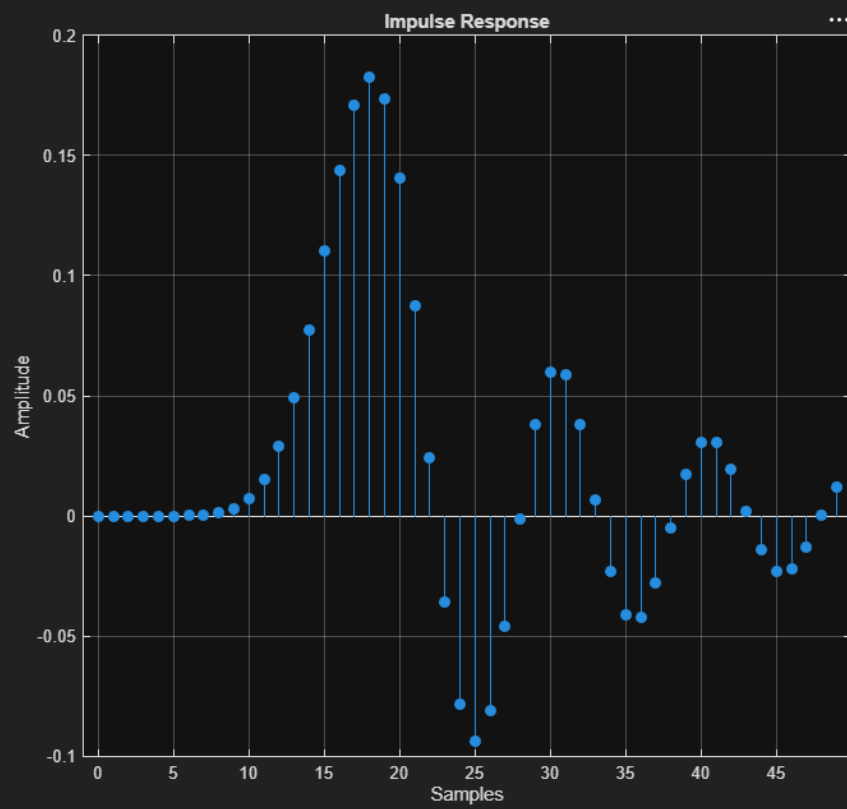
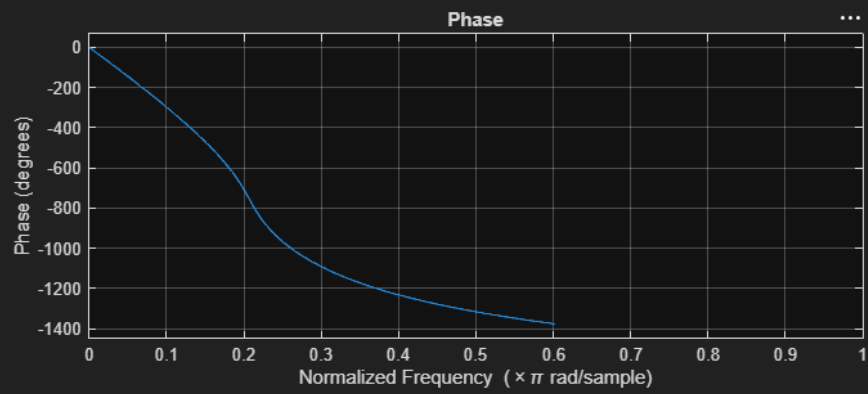
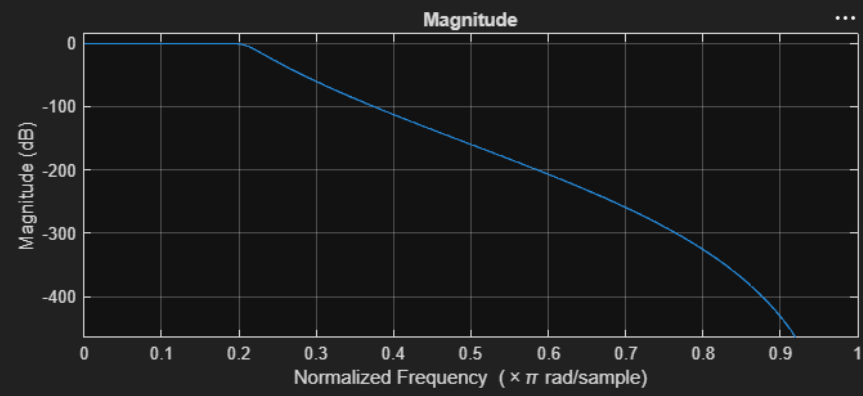
```
1  clc;
2  clear;
3  close all;
4
5
6  Fs = 10000;
7  Fp = 1000;
8  Fst = 1500;
9
10 [N,Wn] = buttord(Fp/(Fs/2),Fst/(Fs/2),1,60);
11 [b,a] = butter(N,Wn);
12
13 % Frequency Response
14 figure;
15 freqz(b,a);
16
17 % Impulse Response
18 [h,n] = impz(b,a,50);
19 figure;
20 stem(n,h,'filled');
21 grid on;
22 xlabel('Samples');
23 ylabel('Amplitude');
24 title('Impulse Response');
25
26 % Step Response
27 figure;
28 stepz(b,a);
29
30 % Pole-Zero Plot
31 figure;
32 zplane(b,a);
33
34 fprintf('Minimum Filter Order (N) = %d\n',N);
35 fprintf('Cutoff Frequency (Wn) = %.4f\n',Wn);
```

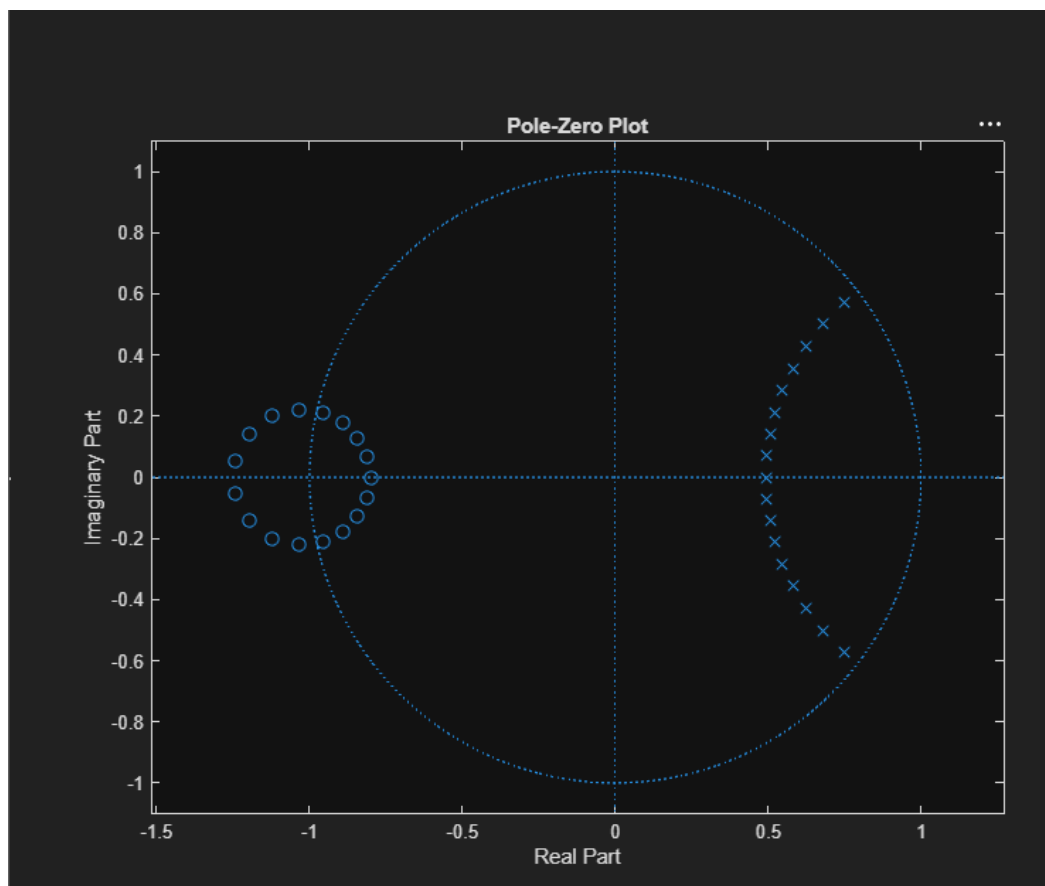
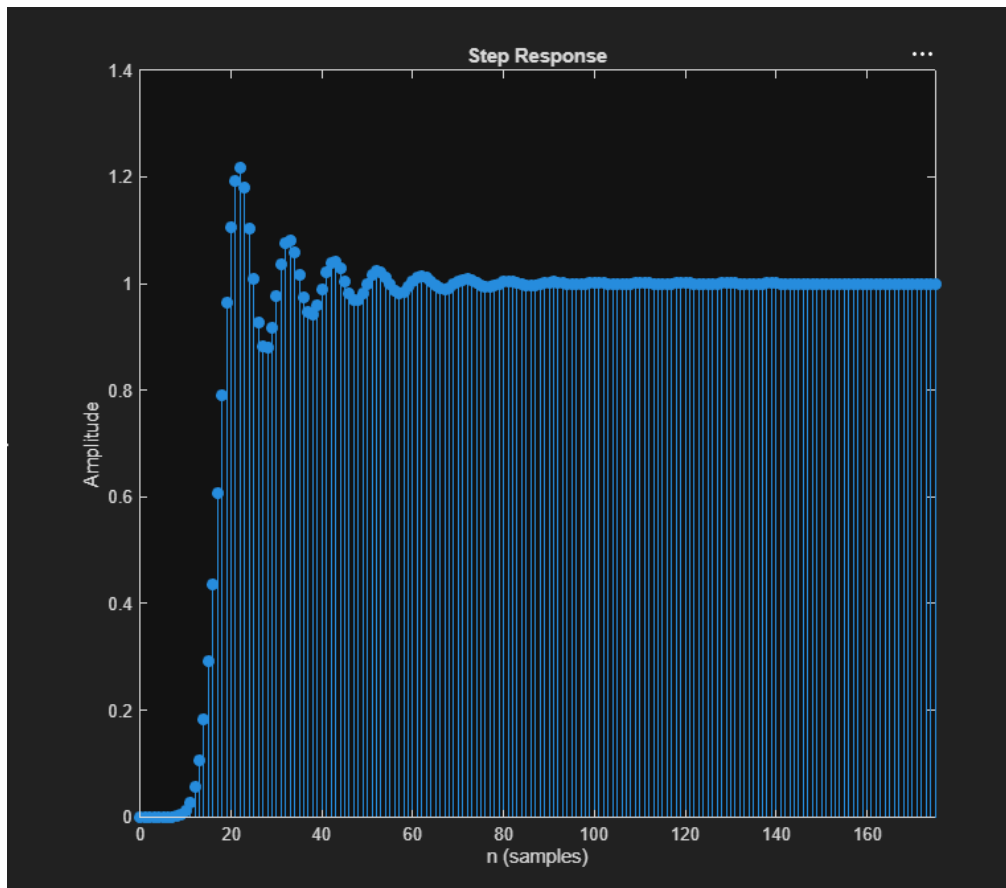
Command Window

New to MATLAB? See resources for [Getting Started](#).

Minimum Filter Order (N) = 17
Cutoff Frequency (Wn) = 0.2083

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1.2 Design and Analysis of a Butterworth IIR High-Pass Filter Using MATLAB

```
1 clc; clear
2
3 Fs=10000; Fp=3000; Fst=2000;
4 [N,Wn]=buttord(Fp/(Fs/2),Fst/(Fs/2),1,60);
5 [b,a]=butter(N,Wn,'high');
6 % Magnitude & Phase
7 freqz(b,a)
8 % Impulse Response
9 figure
10 [h,n]=impz(b,a,50);
11 stem(n,h), grid on
12 title('Impulse Response')
13 % Step Response
14 figure
15 stepz(b,a)
16 title('Step Response')
17 % Group Delay
18 figure
19 grpdelay(b,a)
20 title('Group Delay')
21 % Pole-Zero Plot
22 figure
23 zplane(b,a)
24 title('Pole-Zero Plot')
25 fprintf('Minimum Filter Order (N) = %d\n',N);
26 fprintf('Cutoff Frequency (Wn) = %.4f\n',Wn);
```

Command Window

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Minimum Filter Order (N) = 12
Cutoff Frequency (Wn) = 0.5807

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